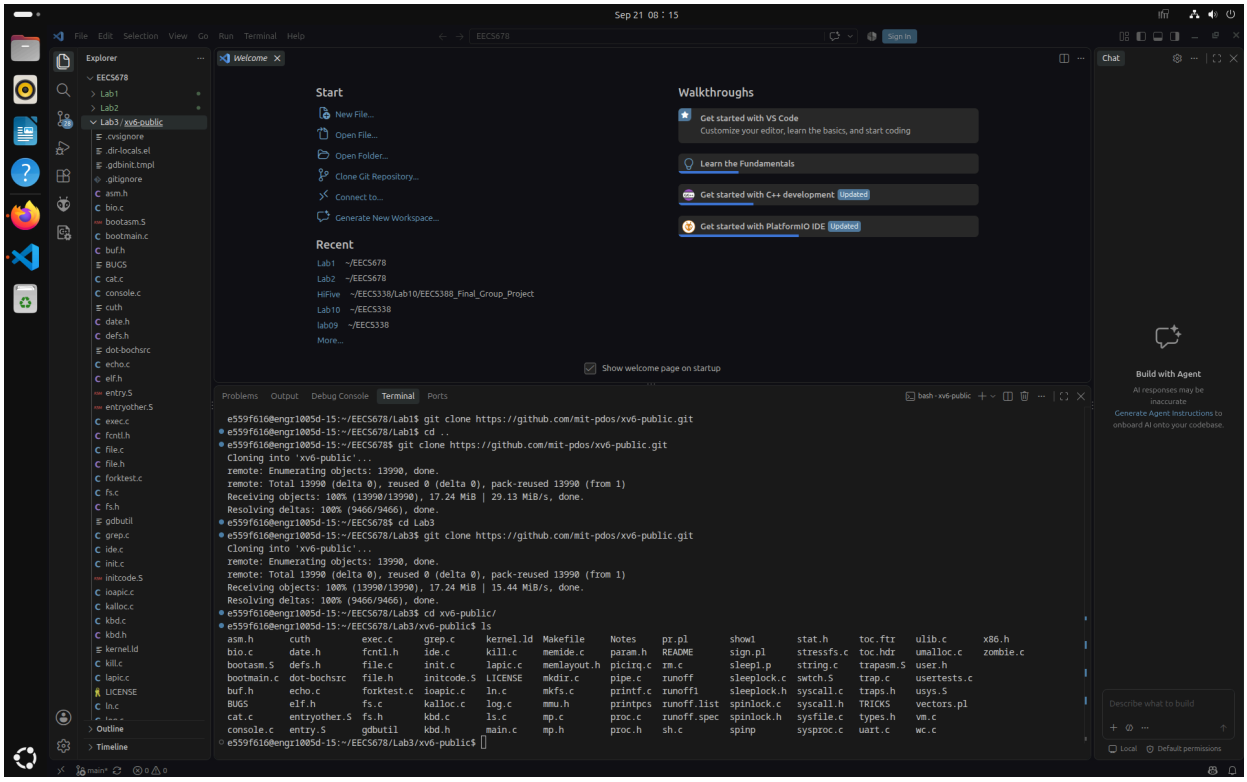
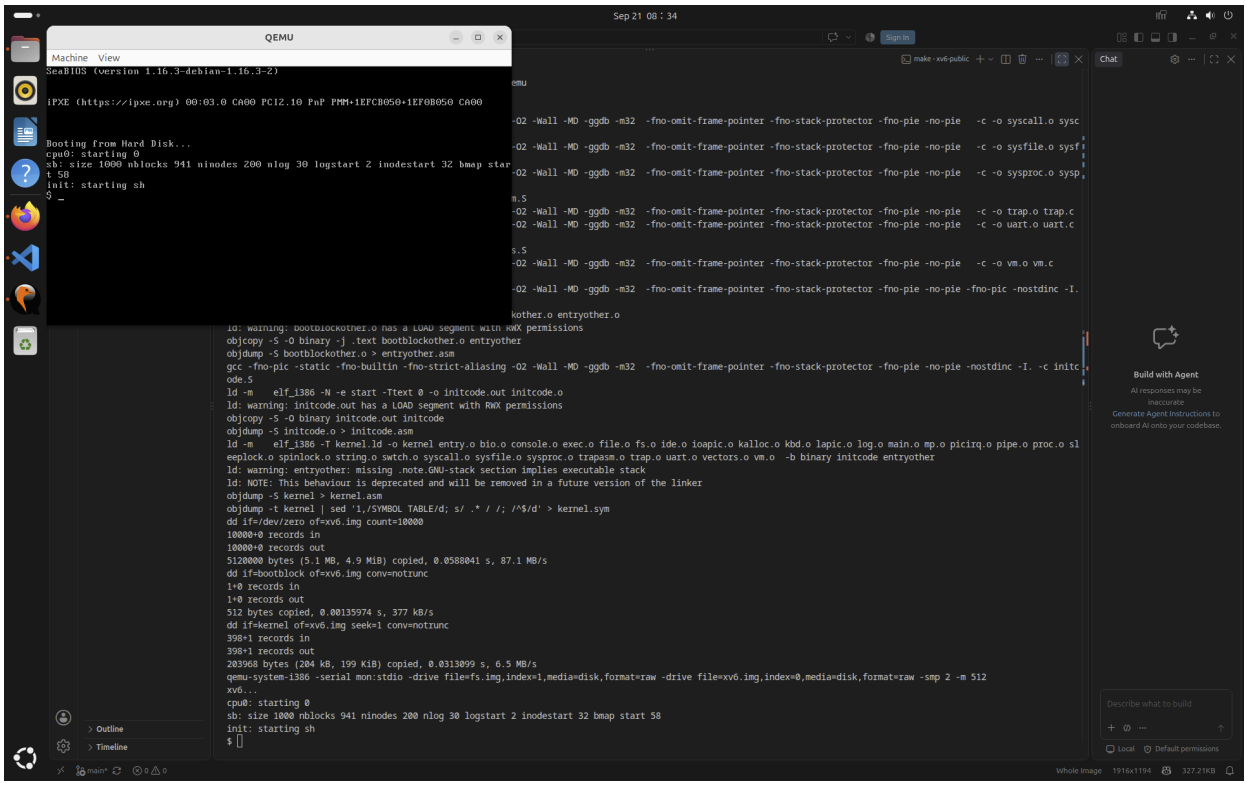
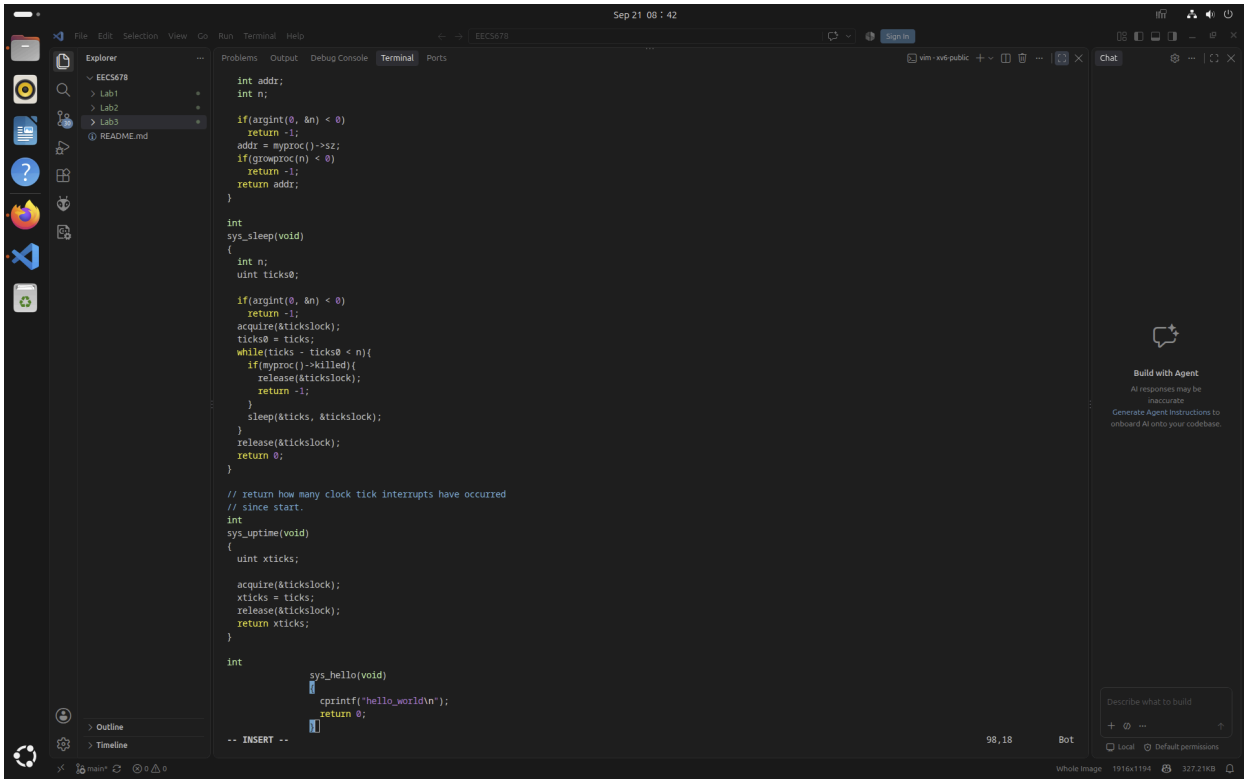
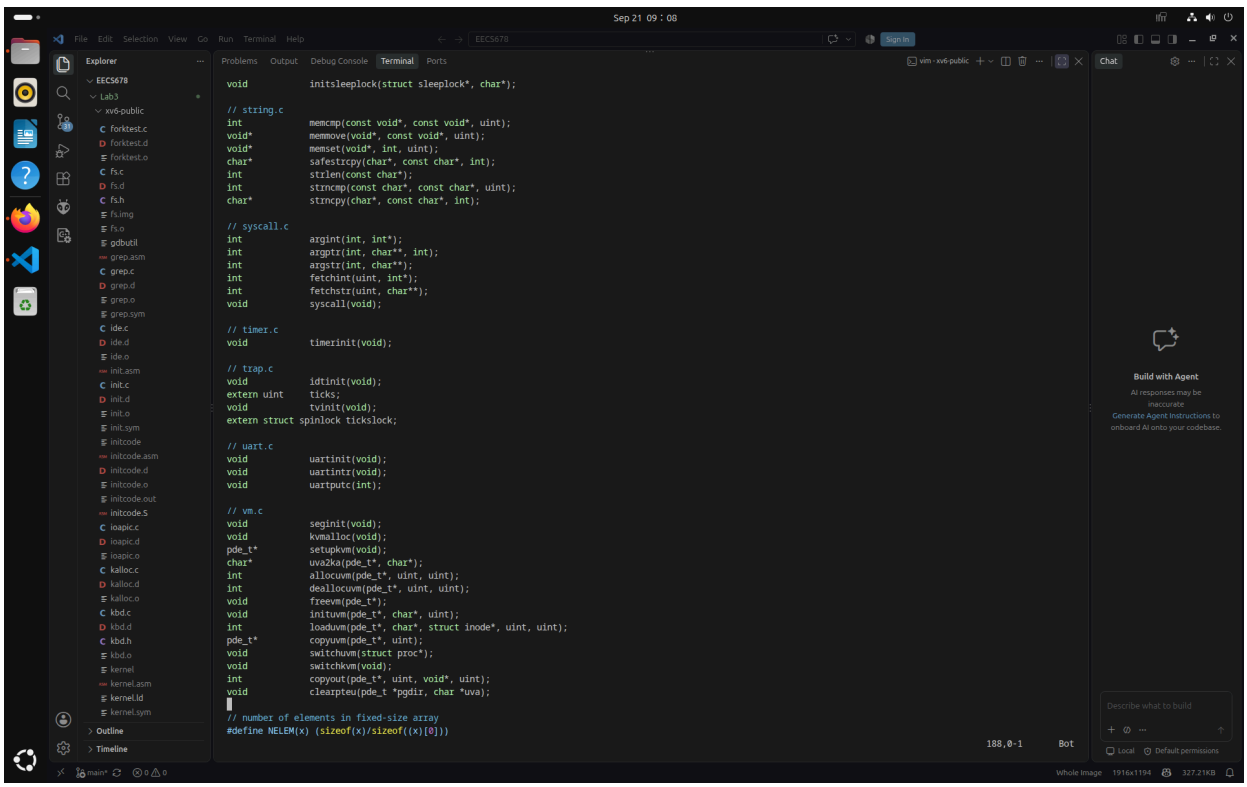
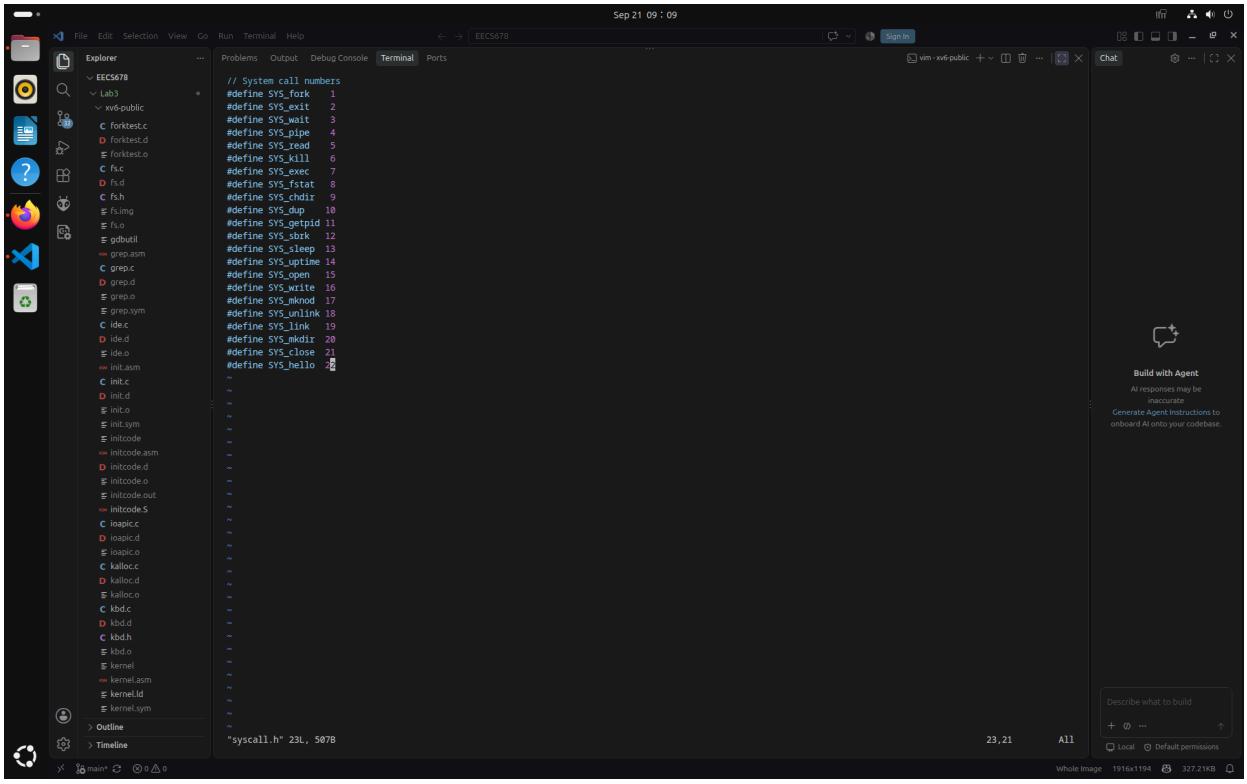
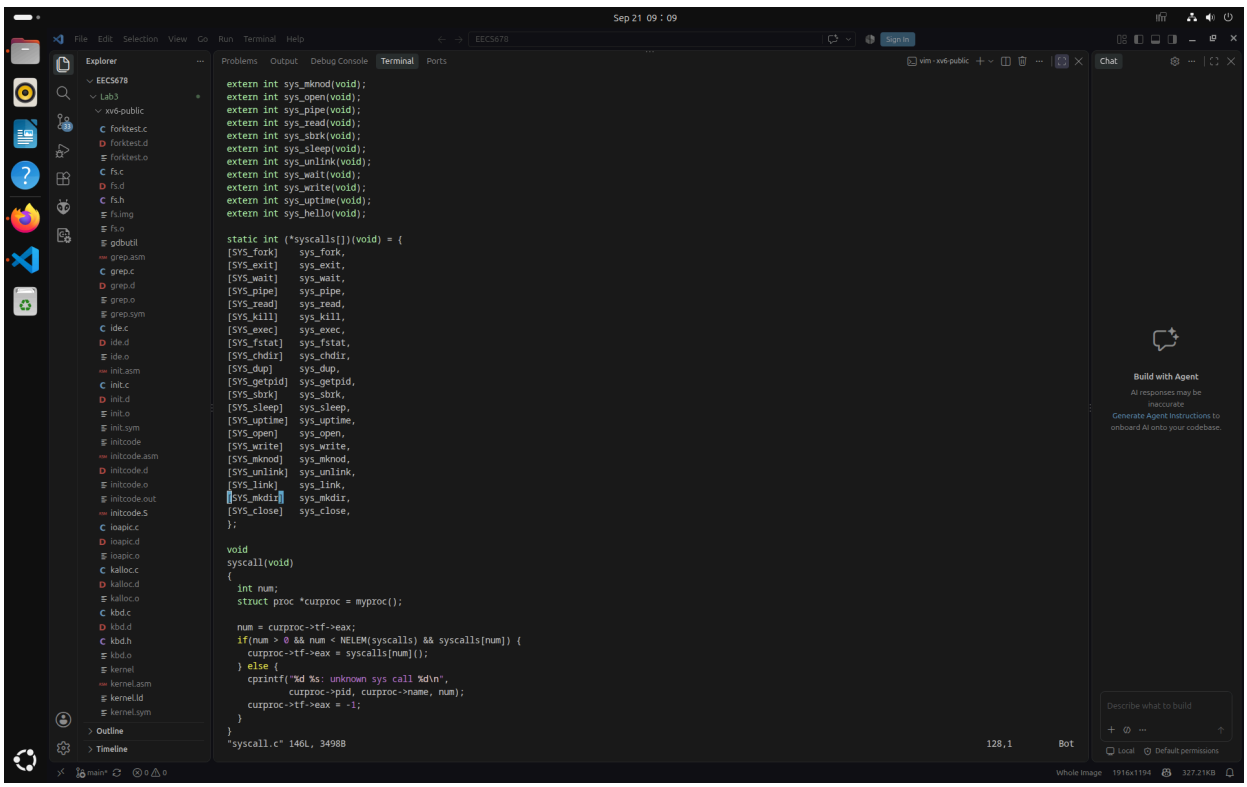

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Start
New File...
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Open Folder...
Clone Git Repository...
Connect to...
Generate New Workspace...
Recent
Lab1 ~/EECS678
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Lab10 ~/EECS338
lab09 ~/EECS338
More...
Walkthroughs
Get started with VS Code
Customize your editor, learn the basics, and start coding
Learn the Fundamentals
Get started with C++ development
Get started with PlatformIO IDE
Show welcome page on startup
e559f616@engr1005d-15:~/EECS678/Lab1$ git clone https://github.com/mit-pdos/xv6-public.git
e559f616@engr1005d-15:~/EECS678/Lab1$ cd ..
e559f616@engr1005d-15:~/EECS678$ git clone https://github.com/mit-pdos/xv6-public.git
Cloning into 'xv6-public'...
remote: Enumerating objects: 13990, done.
remote: Total 13990 (delta 0), reused 0 (delta 0), pack-reused 13990 (from 1)
Receiving objects: 100% (13990/13990), 17.24 MiB | 29.13 MiB/s, done.
Resolving deltas: 100% (9466/9466), done.
e559f616@engr1005d-15:~/EECS678$ cd Lab3
e559f616@engr1005d-15:~/EECS678/Lab3$ git clone https://github.com/mit-pdos/xv6-public.git
Cloning into 'xv6-public'...
remote: Enumerating objects: 13990, done.
remote: Total 13990 (delta 0), reused 0 (delta 0), pack-reused 13990 (from 1)
Receiving objects: 100% (13990/13990), 17.24 MiB | 15.44 MiB/s, done.
Resolving deltas: 100% (9466/9466), done.
e559f616@engr1005d-15:~/EECS678/Lab3$ cd xv6-public/
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$ ls
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$

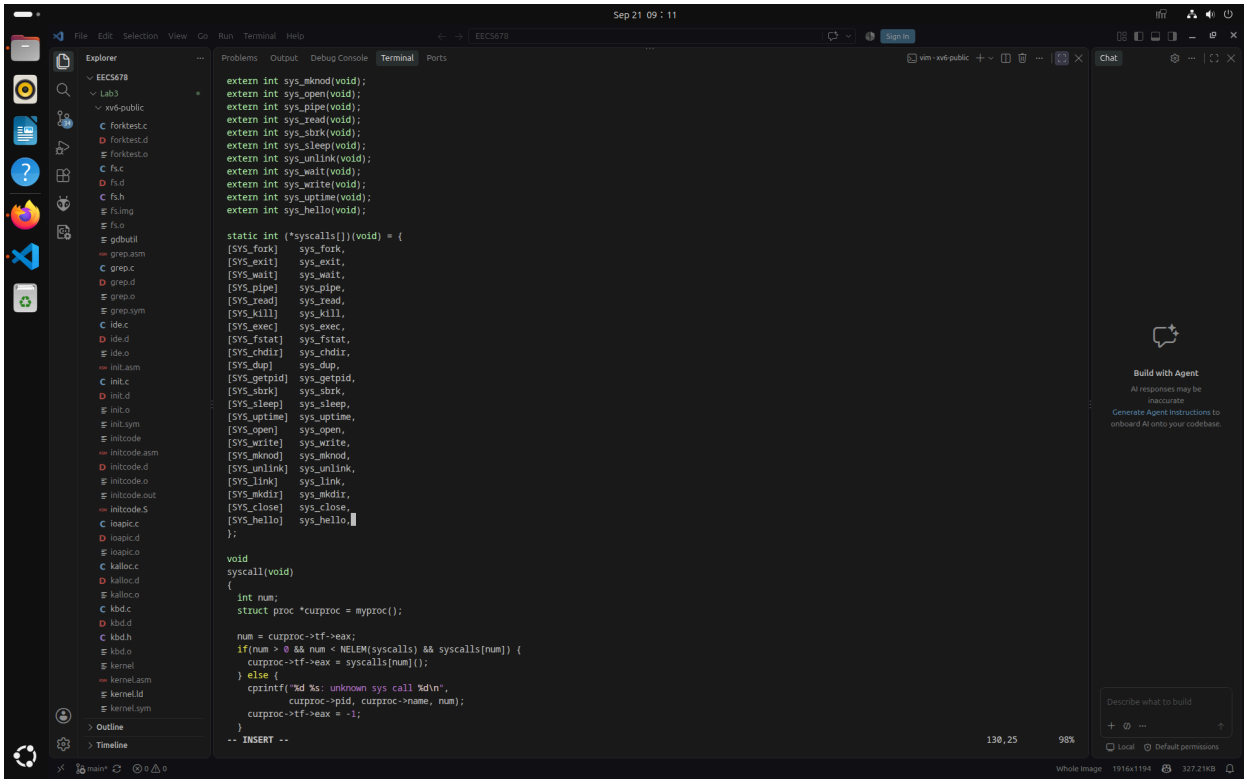

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QEMU
Machine View
SeaBIOS (version 1.16.3-debian-1.16.3-2)
iPXE (https://ipxe.org) 00:03.0 CA00 PCI2.10 PnP PMM+1EFC8050+1EF08050 CA00
Booting from Hard Disk...
cpu0: starting 0
sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 inodestart 32 bmap start 58
init: starting sh
$
objcopy -S -O binary -j .text bootblockother.o entryother
objdump -S bootblockother.o > entryother.asm
gcc -fno-pic -static -fno-builtin -fno-strict-aliasing -O2 -Wall -MD -ggdb -m32 -fno-omit-frame-pointer -fno-stack-protector -fno-pie -no-pie -nostdinc -I. -c initcode.S
ld -m elf_i386 -N -e start -Ttext 0 -o initcode.out initcode.o
ld: warning: initcode.out has a LOAD segment with RWX permissions
objcopy -S -O binary initcode.out initcode
objdump -S initcode.o > initcode.asm
ld -m elf_i386 -T kernel.ld -o kernel entry.o bio.o console.o exec.o file.o fs.o ide.o ioapic.o kalloc.o kbd.o lapic.o log.o main.o mp.o picirq.o pipe.o proc.o sleeplock.o spinlock.o string.o swtch.o syscall.o sysfile.o sysproc.o trapasm.o trap.o uart.o vectors.o vm.o -b binary initcode entryother
ld: warning: entryother: missing .note.GNU-stack section implies executable stack
ld: NOTE: This behaviour is deprecated and will be removed in a future version of the linker
objdump -S kernel > kernel.asm
objdump -t kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel.sym
dd if=/dev/zero of=xv6.img count=10000
10000+0 records in
10000+0 records out
5120000 bytes (5.1 MB, 4.9 MiB) copied, 0.0588041 s, 87.1 MB/s
dd if=bootblock of=xv6.img conv=notrunc
1+0 records in
1+0 records out
512 bytes copied, 0.00135974 s, 377 kB/s
dd if=kernel of=xv6.img seek=1 conv=notrunc
398+1 records in
398+1 records out
203968 bytes (204 kB, 199 KiB) copied, 0.0313099 s, 6.5 MB/s
qemu-system-i386 -serial mon:stdio -drive file=fs.img,index=1,media=disk,format=raw -drive file=xv6.img,index=0,media=disk,format=raw -smp 2 -m 512
xv6...
cpu0: starting 0
sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 inodestart 32 bmap start 58
init: starting sh
$


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int addr;
int n;
if(argint(0, &n) < 0)
return -1;
addr = myproc()->sz;
if(growproc(n) < 0)
return -1;
return addr;
}
int
sys_sleep(void)
{
int n;
uint ticks0;
if(argint(0, &n) < 0)
return -1;
acquire(&tickslock);
ticks0 = ticks;
while(ticks - ticks0 < n){
if(myproc()->killed){
release(&tickslock);
return -1;
}
sleep(&ticks, &tickslock);
}
release(&tickslock);
return 0;
}
// return how many clock tick interrupts have occurred
// since start.
int
sys_uptime(void)
{
uint xticks;
acquire(&tickslock);
xticks = ticks;
release(&tickslock);
return xticks;
}
int
sys_hello(void)
{
cprintf("hello_world\n");
return 0;
}
-- INSERT --
98,18
Bot
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AI responses may be inaccurate
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grep.d
grep.o
grep.sym
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ide.d
ide.o
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init.sym
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initcode.d
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initcode.out
initcode.S
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ioapic.d
ioapic.o
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kalloc.o
kbd.c
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kernel.ld
kernel.sym
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void initsleeplock(struct sleeplock*, char*);
// string.c
int memcmp(const void*, const void*, uint);
void* memmove(void*, const void*, uint);
void* memset(void*, int, uint);
char* safestrcpy(char*, const char*, int);
int strlen(const char*);
int strncmp(const char*, const char*, uint);
char* strncpy(char*, const char*, int);
// syscall.c
int argint(int, int*);
int argptr(int, char**, int);
int argstr(int, char**);
int fetchint(uint, int*);
int fetchstr(uint, char**);
void syscall(void);
// timer.c
void timerinit(void);
// trap.c
void idtinit(void);
extern uint ticks;
void tvinit(void);
extern struct spinlock tickslock;
// uart.c
void uartinit(void);
void uartintr(void);
void uartputc(int);
// vm.c
void seginit(void);
void kvmalloc(void);
pde_t* setupkvm(void);
char* uva2ka(pde_t*, char*);
int allocuvm(pde_t*, uint, uint);
int deallocuvm(pde_t*, uint, uint);
void freevm(pde_t*);
void inituvm(pde_t*, char*, uint);
int loaduvm(pde_t*, char*, struct inode*, uint, uint);
pde_t* copyuvm(pde_t*, uint);
void switchuvm(struct proc*);
void switchkvm(void);
int copyout(pde_t*, uint, void*, uint);
void clearpteu(pde_t *pgdir, char *uva);
// number of elements in fixed-size array
#define NELEM(x) (sizeof(x)/sizeof((x)[0]))
188,0-1
Bot
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ide.d
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init.asm
init.c
init.d
init.o
init.sym
initcode
initcode.asm
initcode.d
initcode.o
initcode.out
initcode.S
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ioapic.d
ioapic.o
kalloc.c
kalloc.d
kalloc.o
kbd.c
kbd.d
kbd.h
kbd.o
kernel
kernel.asm
kernel.ld
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// System call numbers
#define SYS_fork 1
#define SYS_exit 2
#define SYS_wait 3
#define SYS_pipe 4
#define SYS_read 5
#define SYS_kill 6
#define SYS_exec 7
#define SYS_fstat 8
#define SYS_chdir 9
#define SYS_dup 10
#define SYS_getpid 11
#define SYS_sbrk 12
#define SYS_sleep 13
#define SYS_uptime 14
#define SYS_open 15
#define SYS_write 16
#define SYS_mknod 17
#define SYS_unlink 18
#define SYS_link 19
#define SYS_mkdir 20
#define SYS_close 21
#define SYS_hello 22
"syscall.h" 23L, 507B
23,21 All
Chat
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grep.d
grep.o
grep.sym
ide.c
ide.d
ide.o
init.asm
init.c
init.d
init.o
init.sym
initcode
initcode.asm
initcode.d
initcode.o
initcode.out
initcode.S
ioapic.c
ioapic.d
ioapic.o
kalloc.c
kalloc.d
kalloc.o
kbd.c
kbd.d
kbd.h
kbd.o
kernel
kernel.asm
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kernel.sym
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vim - xv6-public
extern int sys_mknod(void);
extern int sys_open(void);
extern int sys_pipe(void);
extern int sys_read(void);
extern int sys_sbrk(void);
extern int sys_sleep(void);
extern int sys_unlink(void);
extern int sys_wait(void);
extern int sys_write(void);
extern int sys_uptime(void);
extern int sys_hello(void);
static int (*syscalls[])(void) = {
[SYS_fork] sys_fork,
[SYS_exit] sys_exit,
[SYS_wait] sys_wait,
[SYS_pipe] sys_pipe,
[SYS_read] sys_read,
[SYS_kill] sys_kill,
[SYS_exec] sys_exec,
[SYS_fstat] sys_fstat,
[SYS_chdir] sys_chdir,
[SYS_dup] sys_dup,
[SYS_getpid] sys_getpid,
[SYS_sbrk] sys_sbrk,
[SYS_sleep] sys_sleep,
[SYS_uptime] sys_uptime,
[SYS_open] sys_open,
[SYS_write] sys_write,
[SYS_mknod] sys_mknod,
[SYS_unlink] sys_unlink,
[SYS_link] sys_link,
[SYS_mkdir] sys_mkdir,
[SYS_close] sys_close,
};
void
syscall(void)
{
int num;
struct proc *curproc = myproc();
num = curproc->tf->eax;
if(num > 0 && num < NELEM(syscalls) && syscalls[num]) {
curproc->tf->eax = syscalls[num]();
} else {
cprintf("%d %s: unknown sys call %d\n",
curproc->pid, curproc->name, num);
curproc->tf->eax = -1;
}
}
"syscall.c" 146L, 3498B
128,1 Bot
Chat
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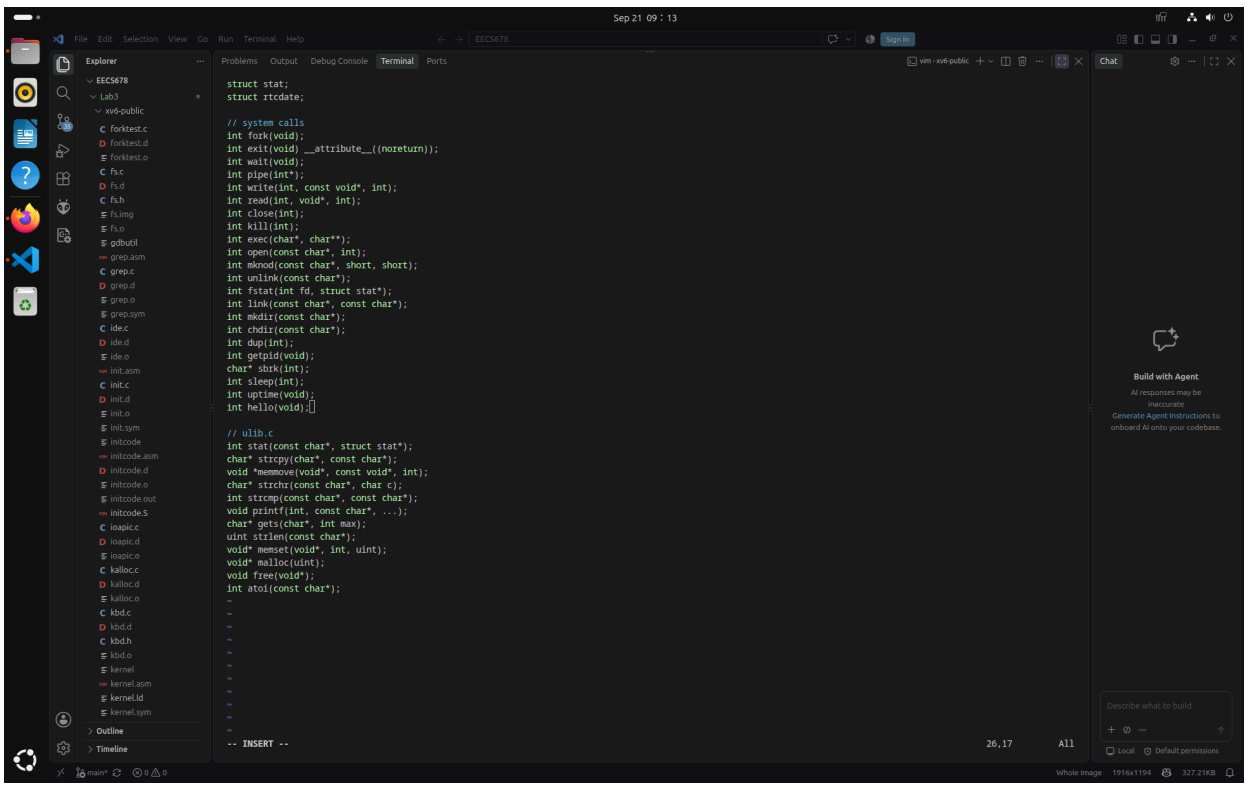



```
extern int sys_mknod(void);
extern int sys_open(void);
extern int sys_pipe(void);
extern int sys_read(void);
extern int sys_sbrk(void);
extern int sys_sleep(void);
extern int sys_unlink(void);
extern int sys_wait(void);
extern int sys_write(void);
extern int sys_uptime(void);
extern int sys_hello(void);

static int (*syscalls[])(void) = {
[SYS_fork]    sys_fork,
[SYS_exit]    sys_exit,
[SYS_wait]    sys_wait,
[SYS_pipe]    sys_pipe,
[SYS_read]    sys_read,
[SYS_kill]    sys_kill,
[SYS_exec]    sys_exec,
[SYS_fstat]   sys_fstat,
[SYS_chdir]   sys_chdir,
[SYS_dup]     sys_dup,
[SYS_getpid]  sys_getpid,
[SYS_sbrk]    sys_sbrk,
[SYS_sleep]   sys_sleep,
[SYS_uptime]  sys_uptime,
[SYS_open]    sys_open,
[SYS_write]   sys_write,
[SYS_mknod]   sys_mknod,
[SYS_unlink]  sys_unlink,
[SYS_link]    sys_link,
[SYS_mkdir]   sys_mkdir,
[SYS_close]   sys_close,
[SYS_hello]   sys_hello,
};

void
syscall(void)
{
  int num;
  struct proc *curproc = myproc();

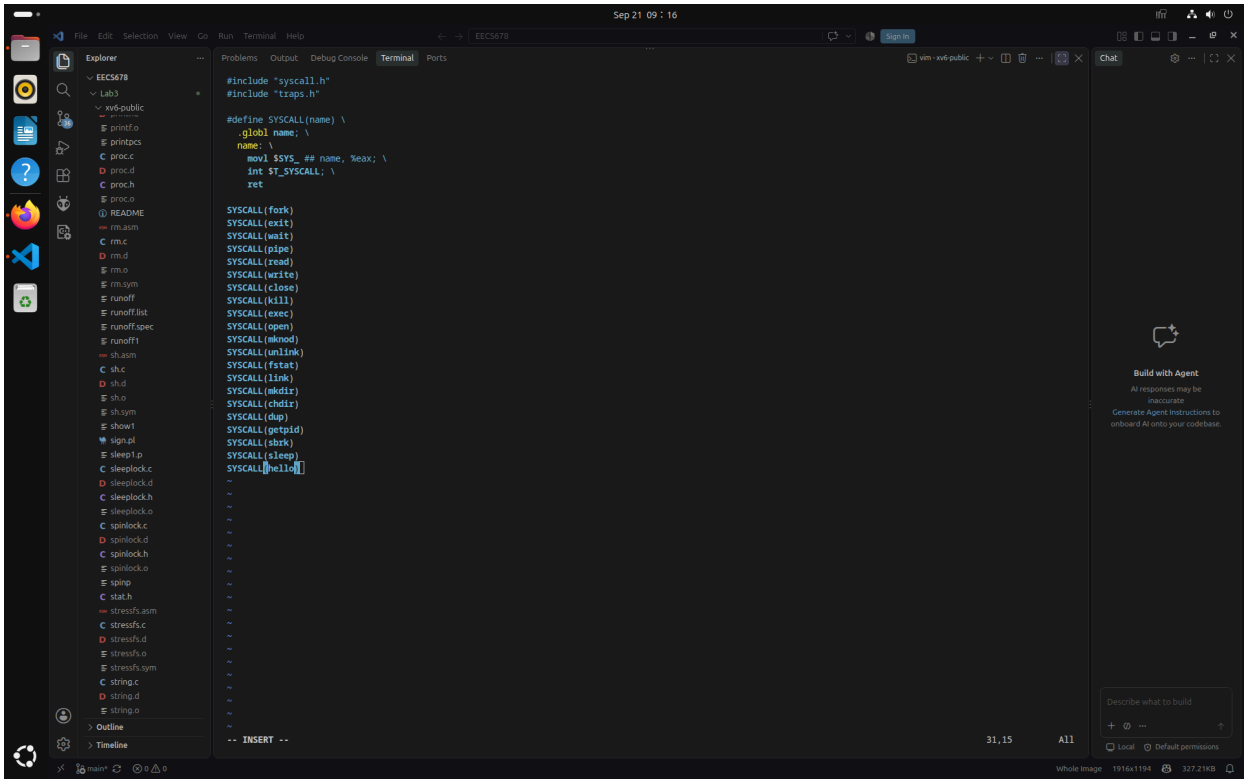
  num = curproc->tf->eax;
  if(num > 0 && num < NELEM(syscalls) && syscalls[num]) {
    curproc->tf->eax = syscalls[num]();
  } else {
    cprintf("%d %s: unknown sys call %d\n",
            curproc->pid, curproc->name, num);
    curproc->tf->eax = -1;
  }
```

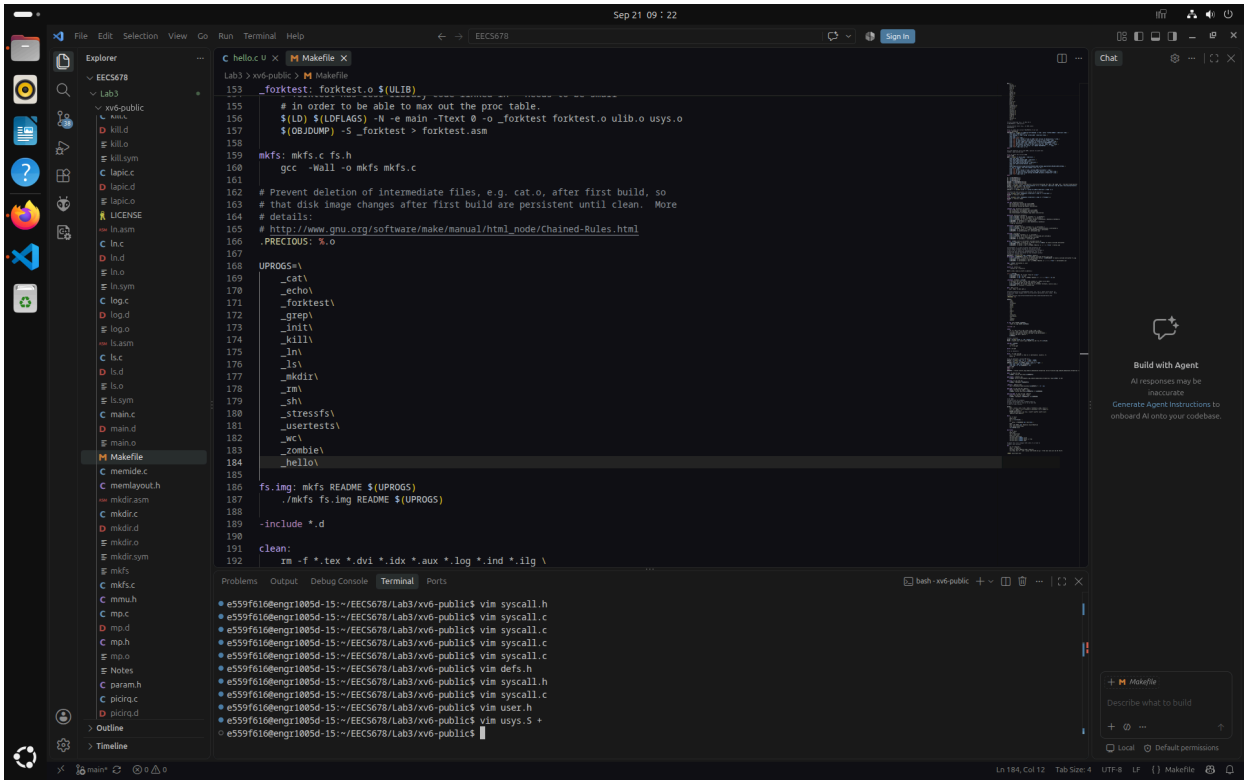


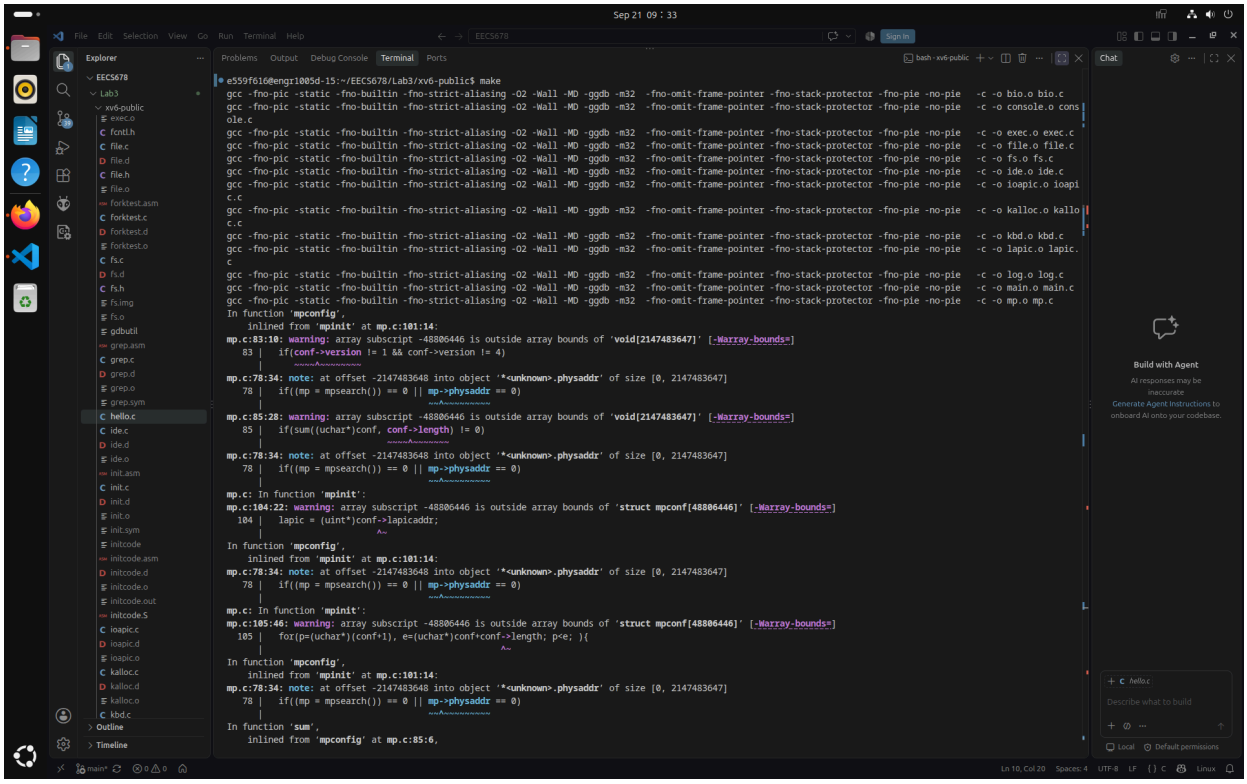

```
struct stat;
struct rtcdate;

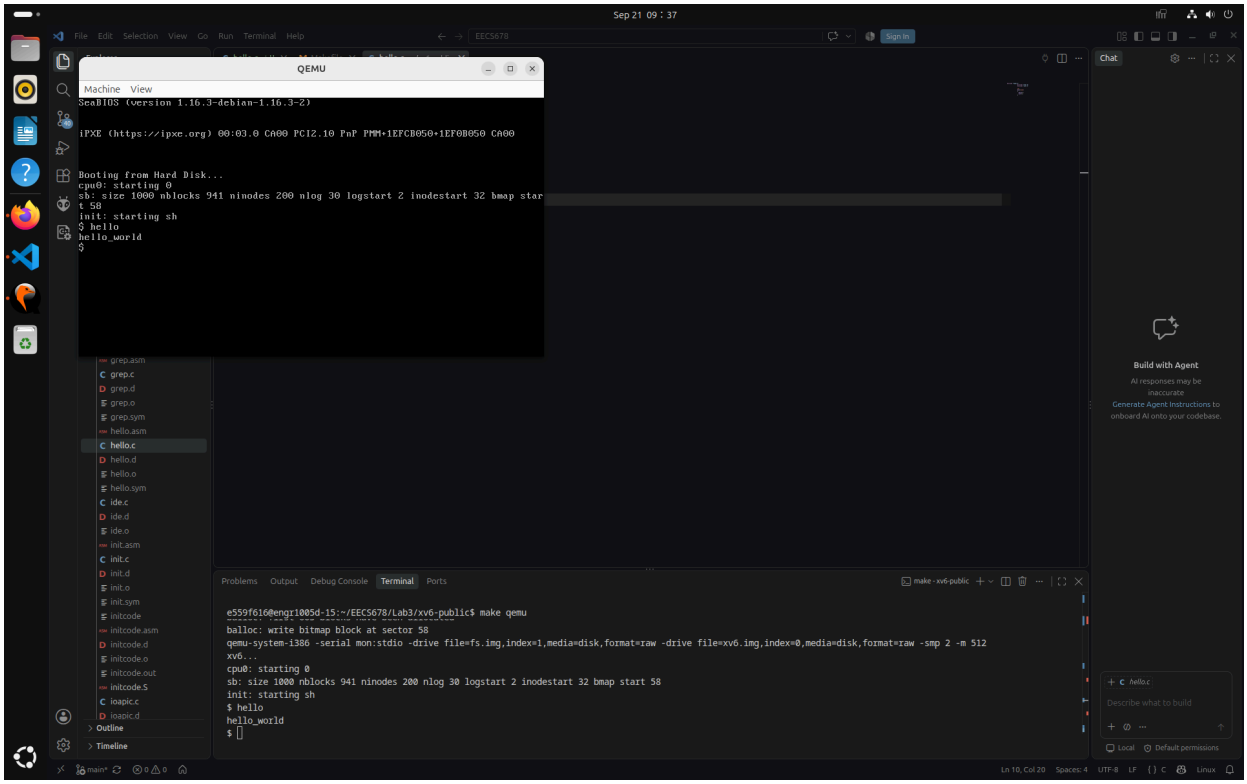
// system calls
int fork(void);
int exit(void) __attribute__((noreturn));
int wait(void);
int pipe(int*);
int write(int, const void*, int);
int read(int, void*, int);
int close(int);
int kill(int);
int exec(char*, char**);
int open(const char*, int);
int mknod(const char*, short, short);
int unlink(const char*);
int fstat(int fd, struct stat*);
int link(const char*, const char*);
int mkdir(const char*);
int chdir(const char*);
int dup(int);
int getpid(void);
char* sbrk(int);
int sleep(int);
int uptime(void);
int hello(void);

// ulib.c
int stat(const char*, struct stat*);
char* strcpy(char*, const char*);
void *memmove(void*, const void*, int);
char* strchr(const char*, char c);
int strcmp(const char*, const char*);
void printf(int, const char*, ...);
char* gets(char*, int max);
uint strlen(const char*);
void* memset(void*, int, uint);
void* malloc(uint);
void free(void*);
int atoi(const char*);
~
~
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~
```


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Chat
#include "syscall.h"
#include "traps.h"
#define SYSCALL(name) \
.globl name; \
name: \
movl $SYS_ ## name, %eax; \
int $T_SYSCALL; \
ret
SYSCALL(fork)
SYSCALL(exit)
SYSCALL(wait)
SYSCALL(pipe)
SYSCALL(read)
SYSCALL(write)
SYSCALL(close)
SYSCALL(kill)
SYSCALL(exec)
SYSCALL(open)
SYSCALL(mknod)
SYSCALL(unlink)
SYSCALL(fstat)
SYSCALL(link)
SYSCALL(mkdir)
SYSCALL(chdir)
SYSCALL(dup)
SYSCALL(getpid)
SYSCALL(sbrk)
SYSCALL(sleep)
SYSCALL(hello)
-- INSERT --
31,15
All
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xv6-public
hello.c U
Makefile
Lab3 > xv6-public > Makefile
_forktest: forktest.o $(ULIB)
in order to be able to max out the proc table.
$(LD) $(LDFLAGS) -N -e main -Ttext 0 -o _forktest forktest.o ulib.o usys.o
$(OBJDUMP) -S _forktest > forktest.asm
mkfs: mkfs.c fs.h
gcc -Wall -o mkfs mkfs.c
Prevent deletion of intermediate files, e.g. cat.o, after first build, so
that disk image changes after first build are persistent until clean. More
details:
http://www.gnu.org/software/make/manual/html_node/Chained-Rules.html
.PRECIOUS: %.o
UPROGS=\
_cat\
_echo\
_forktest\
_grep\
_init\
_kill\
_ln\
_ls\
_mkdir\
_rm\
_sh\
_stressfs\
_usertests\
_wc\
_zombie\
_hello\
fs.img: mkfs README $(UPROGS)
./mkfs fs.img README $(UPROGS)
-include *.d
clean:
rm -f *.tex *.dvi *.idx *.aux *.log *.ind *.ilg \
Problems Output Debug Console Terminal Ports
bash - xv6-public
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$ vim syscall.h
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$ vim syscall.c
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$ vim syscall.c
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$ vim syscall.c
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$ vim syscall.c
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$ vim defs.h
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$ vim syscall.h
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$ vim syscall.c
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$ vim user.h
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$ vim usys.S +
e559f616@engr1005d-15:~/EECS678/Lab3/xv6-public$
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Outline
Timeline
main*
Ln 184, Col 12
Tab Size: 4
UTF-8
LF
Makefile





I got xv6 from the github link and removed -Werror from the make file, and then booted xv6 using QEMU. Then I added a new system call, sysproc.c contained the code that runs when the

system call is invoked, defs.h allowed the other kernel files to call my function, in syscall.h I gave my system call an ID, syscall.c tells the kernel when a user calls my unique ID run sys_hello, user.h contained the function that the user program will call, then i recompiled the kernel using the make file including the new hello.c file I added, and ran the command inside the vm, finally printing hello_world.